

24BDE0045

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School of Computer Science Engineering and Information Systems

BCSE302L- Database Systems

Lab Assessment -5 [PL/ SQL Basics]

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1. Write a simple PL/SQL block to.

a) Print the Fibonacci series.

```
SQL> DECLARE
  2     N NUMBER := &N;
  3     A NUMBER := 0;
  4     B NUMBER := 1;
  5     C NUMBER;
  6 BEGIN
  7     DBMS_OUTPUT.PUT_LINE('FIBONACCI SERIES:');
  8
  9     FOR I IN 1..N LOOP
 10         DBMS_OUTPUT.PUT_LINE(A);
 11
 12         C := A + B;
 13         A := B;
 14         B := C;
 15     END LOOP;
 16 END;
 17 /
Enter value for n: 10
old 2:  N NUMBER := &N;
new 2:  N NUMBER := 10;
FIBONACCI SERIES:
0
1
1
2
3
5
8
13
21
34

PL/SQL procedure successfully completed.
```

b) Print the Factorial of a given number.

```
SQL> DECLARE
 2      N NUMBER := &N;
 3      FACT NUMBER := 1;
 4  BEGIN
 5      FOR I IN 1..N LOOP
 6          FACT := FACT * I;
 7      END LOOP;
 8
 9      DBMS_OUTPUT.PUT_LINE(
10          'FACTORIAL OF ' || N || ' IS: ' || FACT
11      );
12  END;
13  /
Enter value for n: 5
old 2:      N NUMBER := &N;
new 2:      N NUMBER := 5;
FACTORIAL OF 5 IS: 120

PL/SQL procedure successfully completed.
```

c) Print senior citizen based on the age of the customer.

```
SQL> DECLARE
  2     V_CID CUSTOMER.CID%TYPE := &CID;
  3     V_NAME CUSTOMER.NAME%TYPE;
  4     V_DOB CUSTOMER.DOB%TYPE;
  5     V_AGE NUMBER;
  6 BEGIN
  7     SELECT NAME, DOB
  8     INTO V_NAME, V_DOB
  9     FROM CUSTOMER
 10     WHERE CID = V_CID;
 11
 12     V_AGE := TRUNC(MONTHS_BETWEEN(SYSDATE, V_DOB) / 12);
 13
 14     DBMS_OUTPUT.PUT_LINE('CUSTOMER NAME: ' || V_NAME);
 15     DBMS_OUTPUT.PUT_LINE('AGE: ' || V_AGE);
 16
 17     IF V_AGE >= 60 THEN
 18         DBMS_OUTPUT.PUT_LINE('SENIOR CITIZEN: YES');
 19     ELSE
 20         DBMS_OUTPUT.PUT_LINE('SENIOR CITIZEN: NO');
 21     END IF;
 22
 23 EXCEPTION
 24     WHEN NO_DATA_FOUND THEN
 25         DBMS_OUTPUT.PUT_LINE('CUSTOMER NOT FOUND. ');
 26 END;
 27 /
Enter value for cid: 1
old 2:     V_CID CUSTOMER.CID%TYPE := &CID;
new 2:     V_CID CUSTOMER.CID%TYPE := 1;
CUSTOMER NAME: Aarav
AGE: 64
SENIOR CITIZEN: YES

PL/SQL procedure successfully completed.
```

d) Find out the number of customers who have more than 1 accounts.

```
SQL> DECLARE
  2     V_COUNT NUMBER;
  3 BEGIN
  4     SELECT COUNT(*)
  5     INTO V_COUNT
  6     FROM
  7     (
  8         SELECT CID
  9         FROM ACCOUNT
 10        GROUP BY CID
 11        HAVING COUNT(*) > 1
 12     );
 13
 14     DBMS_OUTPUT.PUT_LINE(
 15     'NUMBER OF CUSTOMERS HAVING MORE THAN 1 ACCOUNT: ' || V_COUNT
 16     );
 17 END;
 18 /
NUMBER OF CUSTOMERS HAVING MORE THAN 1 ACCOUNT: 3

PL/SQL procedure successfully completed.
```

e) Display the total number of transactions a customer has made so far.

```
SQL> DECLARE
2     V_CID CUSTOMER.CID%TYPE := &CID;
3     V_COUNT NUMBER;
4 BEGIN
5     SELECT COUNT(*)
6     INTO V_COUNT
7     FROM TRANSACTIONS_WITHINBANK T
8     WHERE T.ID IN
9     (
10        SELECT AID
11        FROM ACCOUNT
12        WHERE CID = V_CID
13
14        UNION
15
16        SELECT LNO
17        FROM LOAN
18        WHERE CID = V_CID
19    );
20
21    DBMS_OUTPUT.PUT_LINE(
22        'TOTAL TRANSACTIONS MADE BY CUSTOMER ' ||
23        V_CID || ': ' || V_COUNT
24    );
25 END;
26 /
Enter value for cid: 1
old 2:     V_CID CUSTOMER.CID%TYPE := &CID;
new 2:     V_CID CUSTOMER.CID%TYPE := 1;
TOTAL TRANSACTIONS MADE BY CUSTOMER 1: 3

PL/SQL procedure successfully completed.
```

f) Find out the number of loans taken by a customer, taking CID as input.

```
SQL> DECLARE
 2     V_CID CUSTOMER.CID%TYPE := &CID;
 3     V_COUNT NUMBER;
 4 BEGIN
 5     SELECT COUNT(*)
 6     INTO V_COUNT
 7     FROM LOAN
 8     WHERE CID = V_CID;
 9
10     DBMS_OUTPUT.PUT_LINE(
11         'NUMBER OF LOANS TAKEN BY CUSTOMER ' ||
12         V_CID || ': ' || V_COUNT
13     );
14 END;
15 /
Enter value for cid: 2
old 2:     V_CID CUSTOMER.CID%TYPE := &CID;
new 2:     V_CID CUSTOMER.CID%TYPE := 2;
NUMBER OF LOANS TAKEN BY CUSTOMER 2: 2

PL/SQL procedure successfully completed.
```

g) Find out the number of customers who do not have any loan.

```
SQL> DECLARE
  2     V_COUNT NUMBER;
  3 BEGIN
  4     SELECT COUNT(*)
  5     INTO V_COUNT
  6     FROM CUSTOMER C
  7     WHERE NOT EXISTS
  8     (
  9         SELECT 1
 10         FROM LOAN L
 11         WHERE L.CID = C.CID
 12     );
 13     DBMS_OUTPUT.PUT_LINE(
 14         'NUMBER OF CUSTOMERS WITHOUT ANY LOAN: ' || V_COUNT
 15     );
 16 END;
 17 /
NUMBER OF CUSTOMERS WITHOUT ANY LOAN: 3

PL/SQL procedure successfully completed.
```


h) Find out the number of customers having both an account and a loan.

```
SQL> DECLARE
  2     V_COUNT NUMBER;
  3 BEGIN
  4     SELECT COUNT(*)
  5     INTO V_COUNT
  6     FROM CUSTOMER C
  7     WHERE EXISTS
  8     (
  9         SELECT 1
 10         FROM ACCOUNT A
 11         WHERE A.CID = C.CID
 12     )
 13     AND EXISTS
 14     (
 15         SELECT 1
 16         FROM LOAN L
 17         WHERE L.CID = C.CID
 18     );
 19
 20     DBMS_OUTPUT.PUT_LINE(
 21         'NUMBER OF CUSTOMERS HAVING BOTH AN ACCOUNT AND A LOAN: ' ||
 22         V_COUNT
 23     );
 24 END;
 25 /
NUMBER OF CUSTOMERS HAVING BOTH AN ACCOUNT AND A LOAN: 6

PL/SQL procedure successfully completed.
```

i) Display the total transaction amount made by a customer, taking CID as input.

```
SQL> DECLARE
  2     V_CID CUSTOMER.CID%TYPE := &CID;
  3     V_TOTAL NUMBER;
  4 BEGIN
  5     SELECT NVL(SUM(T.AMOUNT), 0)
  6     INTO V_TOTAL
  7     FROM TRANSACTIONS_WITHINBANK T
  8     WHERE T.ID IN
  9     (
 10         SELECT AID
 11         FROM ACCOUNT
 12         WHERE CID = V_CID
 13
 14         UNION
 15
 16         SELECT LNO
 17         FROM LOAN
 18         WHERE CID = V_CID
 19     );
 20
 21     DBMS_OUTPUT.PUT_LINE(
 22         'TOTAL TRANSACTION AMOUNT FOR CUSTOMER ' ||
 23         V_CID || ': ' || V_TOTAL
 24     );
 25 END;
 26 /
Enter value for cid: 2
old  2:      V_CID CUSTOMER.CID%TYPE := &CID;
new  2:      V_CID CUSTOMER.CID%TYPE := 2;
TOTAL TRANSACTION AMOUNT FOR CUSTOMER 2: 107000

PL/SQL procedure successfully completed.
```

2. Write a cursor for the following.

a) Display the mini statement for a customer for a particular period given his CID and account or Loan ID.

```
SQL> DECLARE
2  V_CID CUSTOMER.CID%TYPE := &CID;
3  V_ID NUMBER := &ID;
4  V_FROM_DATE DATE := TO_DATE('&FROM_DATE','DD-MM-YYYY');
5  V_TO_DATE DATE := TO_DATE('&TO_DATE','DD-MM-YYYY');
6
7  CURSOR C_MINI IS
8  SELECT TID,
9         TYPE,
10        ID,
11        AMOUNT,
12        DATE_TIME,
13        STATUS
14  FROM TRANSACTIONS_WITHINBANK
15  WHERE ID = V_ID
16        AND DATE_TIME >= V_FROM_DATE
17        AND DATE_TIME < V_TO_DATE + 1
18  AND
19  (
20    ID IN
21    (
22      SELECT AID
23      FROM ACCOUNT
24      WHERE CID = V_CID
25    )
26    OR
27    ID IN
28    (
29      SELECT LNO
30      FROM LOAN
31      WHERE CID = V_CID
32    )
33  )
34  ORDER BY DATE_TIME;
35
36  V_FOUND NUMBER := 0;
37  BEGIN
38    DBMS_OUTPUT.PUT_LINE('MINI STATEMENT');
39    DBMS_OUTPUT.PUT_LINE('-----');
40
41    FOR R IN C_MINI LOOP
42
43      V_FOUND := V_FOUND + 1;
44
45      DBMS_OUTPUT.PUT_LINE(
46        'TID: ' || R.TID ||
47        ' TYPE: ' || R.TYPE ||
48        ' ID: ' || R.ID ||
49        ' AMOUNT: ' || R.AMOUNT ||
50        ' DATE: ' || R.DATE_TIME ||
51        ' STATUS: ' || R.STATUS
52      );
53    END LOOP;
54
55    IF V_FOUND = 0 THEN
56      DBMS_OUTPUT.PUT_LINE('NO TRANSACTIONS FOUND. ');
57    END IF;
58  END;
59 /
Enter value for cid: 1
old 2:  V_CID CUSTOMER.CID%TYPE := &CID;
new 2:  V_CID CUSTOMER.CID%TYPE := 1;
Enter value for id: 101
old 3:  V_ID NUMBER := &ID;
new 3:  V_ID NUMBER := 101;
Enter value for from_date: 01-01-2026
old 4:  V_FROM_DATE DATE := TO_DATE('&FROM_DATE','DD-MM-YYYY');
new 4:  V_FROM_DATE DATE := TO_DATE('01-01-2026','DD-MM-YYYY');
Enter value for to_date: 31-12-2026
old 5:  V_TO_DATE DATE := TO_DATE('&TO_DATE','DD-MM-YYYY');
new 5:  V_TO_DATE DATE := TO_DATE('31-12-2026','DD-MM-YYYY');
MINI STATEMENT
-----
TID: 1 TYPE: D ID: 101 AMOUNT: 5000 DATE: 01-SEP-26 04.56.41.218000 PM STATUS: SUCCESS
TID: 2 TYPE: C ID: 101 AMOUNT: 10000 DATE: 01-SEP-26 04.56.41.222000 PM STATUS: SUCCESS
PL/SQL procedure successfully completed.
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b) Display the bank details like Bank name and all its customer names and city listed.

```
SQL> DECLARE
2   CURSOR C_BANK IS
3   SELECT B.NAME AS BANK_NAME,
4   C.NAME AS CUSTOMER_NAME,
5   C.CITY
6   FROM BANK_MASTER B
7   LEFT JOIN ACCOUNT A
8   ON B.CODE = A.BANK_CODE
9   LEFT JOIN CUSTOMER C
10  ON A.CID = C.CID
11  ORDER BY B.NAME, C.NAME;
12 BEGIN
13   FOR R IN C_BANK LOOP
14   DBMS_OUTPUT.PUT_LINE(
15     'BANK: ' || R.BANK_NAME ||
16     ' | CUSTOMER: ' ||
17     NVL(R.CUSTOMER_NAME, 'NO CUSTOMER') ||
18     ' | CITY: ' ||
19     NVL(R.CITY, 'N/A')
20   );
21   END LOOP;
22 END;
23 /
BANK: HDFC Bank | CUSTOMER: Arjun | CITY: Chennai
BANK: HDFC Bank | CUSTOMER: Sneha | CITY: Chennai
BANK: ICICI Bank | CUSTOMER: Anjali | CITY: Coimbatore
BANK: Indian Overseas Bank | CUSTOMER: Riya | CITY: Bengaluru
BANK: State Bank of India Chennai | CUSTOMER: Akash | CITY: Chennai
BANK: State Bank of India Chennai | CUSTOMER: Maran | CITY: Bengaluru
BANK: State Bank of India Vellore | CUSTOMER: Aarav | CITY: Vellore
BANK: State Bank of India Vellore | CUSTOMER: Akash | CITY: Chennai
BANK: State Bank of India Vellore | CUSTOMER: Rahul | CITY: Vellore
BANK: State Bank of India Vellore | CUSTOMER: Riya | CITY: Bengaluru
BANK: State Bank of India Vellore | CUSTOMER: Sneha | CITY: Chennai
BANK: State Bank of India Vellore | CUSTOMER: Suresh | CITY: Vellore
PL/SQL procedure successfully completed.
```

c) Write a cursor to display the following customer account summary by taking CID as input:

- Customer Name
- Number of Accounts
- Total Account Balance
- Number of Loans
- Total Outstanding Loan Balance

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```
SQL> DECLARE
2     V_CID CUSTOMER.CID%TYPE := &CID;
3
4     V_NAME CUSTOMER.NAME%TYPE;
5     V_ACCOUNT_COUNT NUMBER;
6     V_ACCOUNT_BALANCE NUMBER;
7     V_LOAN_COUNT NUMBER;
8     V_LOAN_BALANCE NUMBER;
9
10    CURSOR C_SUMMARY IS
11        SELECT C.NAME,
12
13            (SELECT COUNT(*)
14             FROM ACCOUNT A
15             WHERE A.CID = C.CID),
16
17            (SELECT NVL(SUM(A.BALANCE),0)
18             FROM ACCOUNT A
19             WHERE A.CID = C.CID),
20
21            (SELECT COUNT(*)
22             FROM LOAN L
23             WHERE L.CID = C.CID),
24
25            (SELECT NVL(SUM(L.BALANCE),0)
26             FROM LOAN L
27             WHERE L.CID = C.CID)
28
29        FROM CUSTOMER C
30        WHERE C.CID = V_CID;
31
32 BEGIN
33     OPEN C_SUMMARY;
34
35     FETCH C_SUMMARY
36     INTO V_NAME,
37         V_ACCOUNT_COUNT,
38         V_ACCOUNT_BALANCE,
39         V_LOAN_COUNT,
40         V_LOAN_BALANCE;
41
42     IF C_SUMMARY%FOUND THEN
43
44         DBMS_OUTPUT.PUT_LINE(
45             'CUSTOMER NAME: ' || V_NAME
46         );
47
48         DBMS_OUTPUT.PUT_LINE(
49             'NUMBER OF ACCOUNTS: ' || V_ACCOUNT_COUNT
50         );
51
52         DBMS_OUTPUT.PUT_LINE(
53             'TOTAL ACCOUNT BALANCE: ' || V_ACCOUNT_BALANCE
54         );
55
56         DBMS_OUTPUT.PUT_LINE(
57             'NUMBER OF LOANS: ' || V_LOAN_COUNT
58         );
59
60         DBMS_OUTPUT.PUT_LINE(
61             'TOTAL OUTSTANDING LOAN BALANCE: ' ||
62             V_LOAN_BALANCE
63         );
64
65     ELSE
66         DBMS_OUTPUT.PUT_LINE('CUSTOMER NOT FOUND. ');
67     END IF;
68
69     CLOSE C_SUMMARY;
70 END;
71 /
Enter value for cid: 1
old 2:  V_CID CUSTOMER.CID%TYPE := &CID;
new 2:  V_CID CUSTOMER.CID%TYPE := 1;
CUSTOMER NAME: Aarav
NUMBER OF ACCOUNTS: 1
TOTAL ACCOUNT BALANCE: 150000
NUMBER OF LOANS: 1
TOTAL OUTSTANDING LOAN BALANCE: 450000
PL/SQL procedure successfully completed.
```

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d) Display the customers who have both an account and a loan in the bank. Display CID, customer name, account ID, and loan number.

```
SQL> DECLARE
2   CURSOR C_CUSTOMER_DETAILS IS
3   SELECT C.CID,
4   C.NAME,
5   A.AID,
6   L.LNO
7   FROM CUSTOMER C
8   JOIN ACCOUNT A
9   ON C.CID = A.CID
10  JOIN LOAN L
11  ON C.CID = L.CID
12  ORDER BY C.CID, A.AID, L.LNO;
13 BEGIN
14   FOR R IN C_CUSTOMER_DETAILS LOOP
15
16     DBMS_OUTPUT.PUT_LINE(
17       'CID: ' || R.CID ||
18       ' NAME: ' || R.NAME ||
19       ' ACCOUNT ID: ' || R.AID ||
20       ' LOAN NO: ' || R.LNO
21     );
22
23   END LOOP;
24 END;
25 /
CID: 1 | NAME: Arav | ACCOUNT ID: 101 | LOAN NO: 1
CID: 2 | NAME: Akash | ACCOUNT ID: 102 | LOAN NO: 2
CID: 2 | NAME: Akash | ACCOUNT ID: 102 | LOAN NO: 7
CID: 2 | NAME: Akash | ACCOUNT ID: 111 | LOAN NO: 2
CID: 2 | NAME: Akash | ACCOUNT ID: 111 | LOAN NO: 7
CID: 3 | NAME: Riya | ACCOUNT ID: 103 | LOAN NO: 3
CID: 3 | NAME: Riya | ACCOUNT ID: 112 | LOAN NO: 3
CID: 4 | NAME: Sneha | ACCOUNT ID: 104 | LOAN NO: 4
CID: 4 | NAME: Sneha | ACCOUNT ID: 104 | LOAN NO: 8
CID: 4 | NAME: Sneha | ACCOUNT ID: 113 | LOAN NO: 4
CID: 4 | NAME: Sneha | ACCOUNT ID: 113 | LOAN NO: 8
CID: 5 | NAME: Rahul | ACCOUNT ID: 105 | LOAN NO: 5
CID: 6 | NAME: Anjali | ACCOUNT ID: 106 | LOAN NO: 6

PL/SQL procedure successfully completed.
```

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e) Display the transaction details of a given account for a specified date range. Take AID, From Date, and To Date as input.

```
SQL> DECLARE
2     V_AID ACCOUNT.AID%TYPE := &AID;
3
4     V_FROM_DATE DATE :=
5         TO_DATE('&FROM_DATE','DD-MM-YYYY');
6
7     V_TO_DATE DATE :=
8         TO_DATE('&TO_DATE','DD-MM-YYYY');
9
10    CURSOR C_TRANSACTION IS
11        SELECT TID,
12               TYPE,
13               ID,
14               AMOUNT,
15               DATE_TIME,
16               STATUS
17        FROM TRANSACTIONS_WITHINBANK
18        WHERE ID = V_AID
19        AND DATE_TIME >= V_FROM_DATE
20        AND DATE_TIME < V_TO_DATE + 1
21        ORDER BY DATE_TIME;
22
23    V_FOUND NUMBER := 0;
24
25 BEGIN
26     DBMS_OUTPUT.PUT_LINE('TRANSACTION DETAILS');
27     DBMS_OUTPUT.PUT_LINE('-----');
28
29     FOR R IN C_TRANSACTION LOOP
30
31         V_FOUND := V_FOUND + 1;
32
33         DBMS_OUTPUT.PUT_LINE(
34             'TID: ' || R.TID ||
35             ' | TYPE: ' || R.TYPE ||
36             ' | ACCOUNT ID: ' || R.ID ||
37             ' | AMOUNT: ' || R.AMOUNT ||
38             ' | DATE: ' || R.DATE_TIME ||
39             ' | STATUS: ' || R.STATUS
40         );
41     END LOOP;
42
43     IF V_FOUND = 0 THEN
44         DBMS_OUTPUT.PUT_LINE(
45             'NO TRANSACTIONS FOUND FOR THE GIVEN PERIOD.'
46         );
47     END IF;
48 END;
49 /
Enter value for aid: 3
old 2:   V_AID ACCOUNT.AID%TYPE := &AID;
new 2:   V_AID ACCOUNT.AID%TYPE := 3;
Enter value for from_date: 01-01-2026
old 5:   TO_DATE('&FROM_DATE','DD-MM-YYYY');
new 5:   TO_DATE('01-01-2026','DD-MM-YYYY');
Enter value for to_date: 31-12-2026
old 8:   TO_DATE('&TO_DATE','DD-MM-YYYY');
new 8:   TO_DATE('31-12-2026','DD-MM-YYYY');
TRANSACTION DETAILS
-----
TID: 11 | TYPE: C | ACCOUNT ID: 3 | AMOUNT: 40000 | DATE: 07-AUG-26 12.00.000000 PM | STATUS: SUCCESS
PL/SQL procedure successfully completed.
```